

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Agentic AI Applications with Claude Code

TGS Ref No: TGS-2025052468

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.2

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	27 Jun 2026	First version — full-day (9:00 AM–6:00 PM) lesson plan	Dr. Alfred Ang
1.1	27 Jun 2026	One-day schedule, 1-hour assessment at 5:00 PM (30 min WA + 30 min PP), LMS	Tertiary Infotech Academy Pte Ltd
1.2	27 Jun 2026	Aligned Topic 1 to the 7-step build workflow and added the Hooks activity	Tertiary Infotech Academy Pte Ltd

TABLE OF CONTENTS

Learning Outcomes

Daily Schedule (9:00 AM – 6:00 PM, one-day class)

Topic-by-Topic Breakdown

Topic 1 — Claude Code Fundamentals ($\approx$ 1 hr 15 min + Activity 1)

Topic 2 — Tools and Commands ($\approx$ 1 hr 30 min incl. activities)

Topic 3 — Skills, Agents & Hooks ($\approx$ 1 hr 30 min + Mini Capstone)

Resources Required

Assessment

Learning Outcomes

By the end of this course, learners will be able to:

1. Explain the agentic loop and apply context-engineering techniques in Claude Code.
 2. Create custom slash commands and connect MCP tools to extend Claude Code.
 3. Use agent skills, sub agents and hooks to automate specialised tasks.
 4. Build, deploy, test, secure and optimise a real website end-to-end with Claude Code.
-

Daily Schedule (9:00 AM – 6:00 PM, one-day class)

This is a **one-day class**, so the **1-hour assessment runs from 5:00 – 6:00 PM** and all teaching content finishes by 5:00 PM. Breaks are shortened (10-minute breaks, 45-minute lunch) to make room.

Time	Duration	Topic / Activity	Method
9:00 – 9:30	30 min	Welcome & Administration — Digital attendance (AM), trainer introduction, learner introductions, ground rules, course outline	Lecture / Discussion
9:30 – 10:30	60 min	Topic 1.1 Claude Code fundamentals — Setup (Terminal, VS Code, Desktop); how it works (agentic loop + built-in tools); permission modes & bypass	Lecture / Demo
10:30 – 10:40	10 min	Morning Break	—
10:40 – 11:20	40 min	Topic 1.2 Context, memory & sessions — context window & /context, CLAUDE.md vs auto memory, managing sessions, scheduled tasks, /goal	Lecture / Demo
11:20 – 12:30	70 min	Activity 1 — Build & Deploy a Website — Build a responsive software-dev site (hero, testimonials, enquiry form), push to GitHub, deploy on GitHub Pages via Actions, add README	Hands-on
12:30 – 1:15	45 min	Lunch Break	—
1:15 – 1:45	30 min	Topic 2.1 Custom Commands — Slash commands, .claude/commands, arguments; Activity 2 — Create /gitpush command	Lecture / Hands-on
1:45 – 2:45	60 min	Topic 2.2 MCP Tools — MCP architecture, connecting servers & scopes, Playwright MCP; Activity 3 —	Lecture / Hands-on

		screenshot site & push; Activity 3B — test enquiry form to email	
2:45 – 2:55	10 min	Afternoon Break	—
2:55 – 3:40	45 min	Topic 3.1 Agent Skills — SKILL.md, project skills, installing skills; Activity 4 — Install frontend-design & seo-audit skills, optimise site	Lecture / Hands-on
3:40 – 4:25	45 min	Topic 3.2 Sub Agents & 3.3 Hooks — Creating sub agents, hook events; Activity 5 — Security sub agent + lead-magnet UI/UX sub agent	Lecture / Hands-on
4:25 – 4:50	25 min	Mini Capstone — Activity 6 & Presentations — Form test to email, WhatsApp widget, final review; learners present	Hands-on / Presentation
4:50 – 5:00	10 min	Summary, Q&A & Closing — Recap, digital attendance (PM), certification & TRAQOM survey	Discussion
5:00 – 5:30	30 min	Written Assessment (WA) — open book, on the LMS	Assessment
5:30 – 6:00	30 min	Practical Performance (PP) — open book, on the LMS	Assessment

The 1-hour assessment (30 min WA + 30 min PP) runs from 5:00 PM and the class ends at 6:00 PM. Breaks are shortened (10-minute breaks, 45-minute lunch) to fit the assessment. Courseware and the assessment are available on the LMS at <https://lms-tms.tertiaryinfotech.com/>. Timings may be adjusted slightly to suit class pace.

Topic-by-Topic Breakdown

Topic 1 — Claude Code Fundamentals (≈ 1 hr 15 min + Activity 1)

- Setup: Terminal, IDE (VS Code) and Desktop
- How Claude Code works: the agentic loop and built-in tools
- Context engineering & the context window (/context, /compact, /clear)
- Memory files: CLAUDE.md vs auto memory (/init, /memory)
- Permission modes and the dangers of bypass permissions
- Managing sessions, scheduled tasks (Routines), and /goal
- **Activity 1:** Build and deploy a website

Topic 2 — Tools and Commands (≈ 1 hr 30 min incl. activities)

- Custom slash commands and arguments
- **Activity 2:** Create a /gitpush custom command
- MCP tools and architecture; Playwright MCP

- **Activity 3:** Install Playwright MCP, screenshot site, push
- **Activity 3B:** Test the enquiry form with Playwright; confirm enquiries reach your email

Topic 3 — Skills, Agents & Hooks (≈ 1 hr 30 min + Mini Capstone)

- Agent skills and installation
 - **Activity 4:** Install & run frontend-design and seo-audit skills
 - Sub agents and hooks
 - **Activity 5:** Security and UI/UX (lead-magnet) sub agents
 - **Activity 6 / Mini Capstone:** Form testing, WhatsApp widget, final review and presentation
-

Resources Required

- Computers with internet access (per learner)
- Claude Code installed and signed in
- GitHub accounts; Node.js installed
- Projector / shared screen for demonstrations
- Course slides and this learner guide

Assessment

- **Written Assessment (WA):** 30 min — open book, on the LMS
 - **Practical Performance (PP):** 30 min — open book, on the LMS
 - Assessment starts at 5:00 PM and the class ends at 6:00 PM.
 - **Courseware & assessment:** available on the LMS at <https://lms-tms.tertiaryinfotech.com/>
 - **Funding criteria:** minimum 75% attendance and assessed as *Competent*
-

Trainer contact: enquiry@tertiaryinfotech.com | +65 6100 0613